

Sizing Grid Operating Reserves Against Tail Risk

A quantitative lab experiment evaluating GP + soft-EM regime-mixture against real ISO practices.

Target	Test Environments	Compute
99.9726% Daily Reliability	Synthetic Oracle (100 sites)	Rust & Vectorized NumPy
	Real EIA-930 Data (15 US Balancing Authorities)	

GP + soft-EM achieves higher out-of-sample reliability at a lower total cost.

Phase 2 Results | Real EIA-930 Data | Fit on 2023 | Scored on Held-Out 2024

METHOD	RESERVE (MW)	ACHIEVED RELIABILITY	TOTAL ANNUAL COST
ERCOT's real N-1 rule	29,610 MW	99.18% Achieved	\$8.42B
Aggregate correlation (real ISO practice)	27,594 MW	99.18% Achieved	\$5.27B
GP + soft-EM regime-mixture	37,215 MW	99.73% Achieved	\$4.95B

Takeaway Callout: The novel regime-mixture approach is the only method that safely approaches the reliability target while simultaneously lowering the billion-dollar total annual cost.

The Sizing Challenge: Translating “1-in-10 ‘Years’” to a Daily Target

Industry Convention



Grid operators size operating reserves against a rigid reliability target. The North American convention (confirmed across six independent regions) is a 0.1 days/year loss-of-load expectation.

Mathematical Target

Translated to a daily target (the replicate unit of this lab), the requirement becomes:

$$\text{target reliability} = 1 - \frac{0.1}{365} = 0.999726$$

Note: For a single aggregate quantity (total fleet-wide shortfall), the Rockafellar-Uryasev CVaR minimization has a closed form: the optimal threshold is the Value-at-Risk at the target quantile (q).

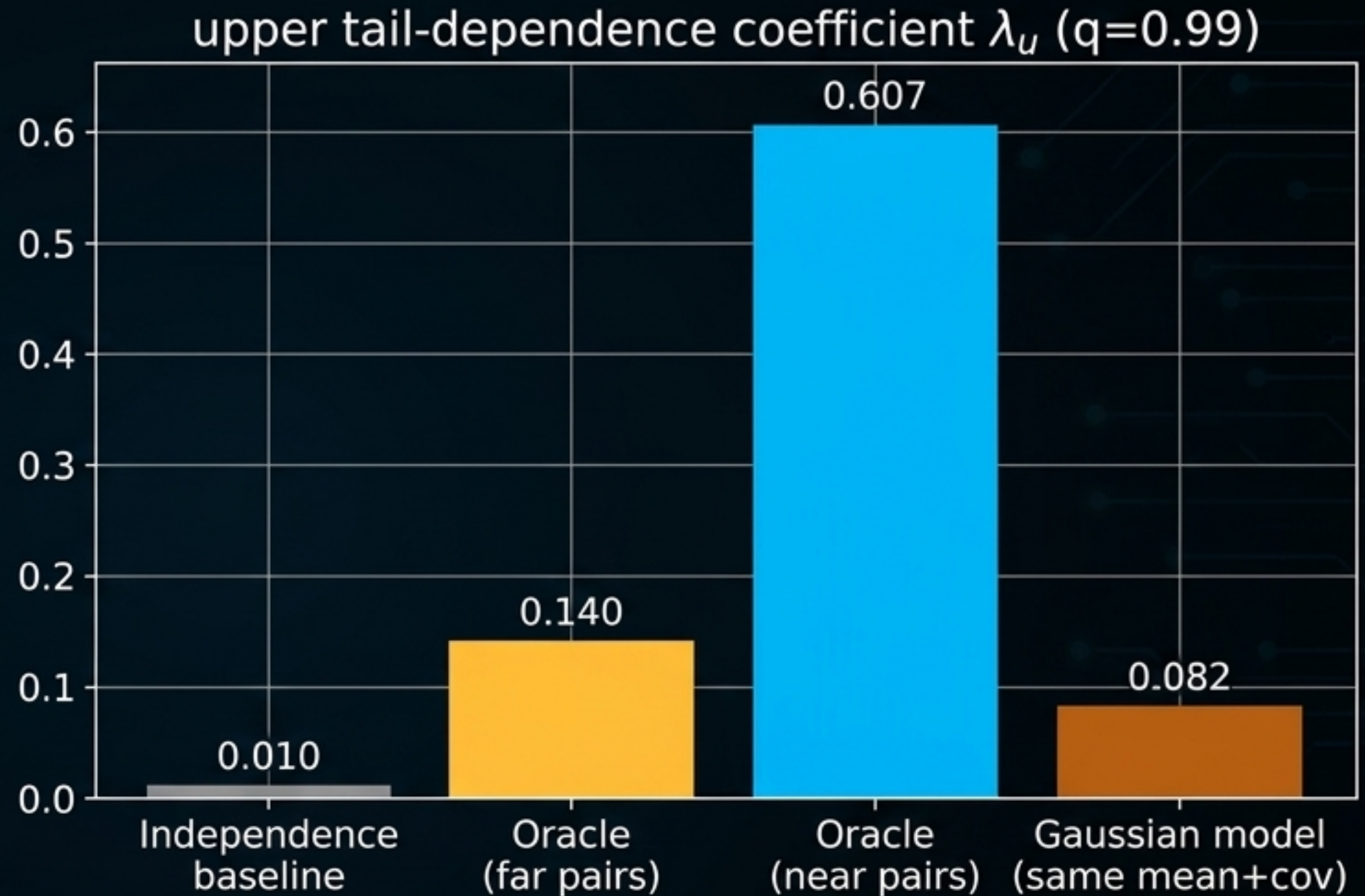
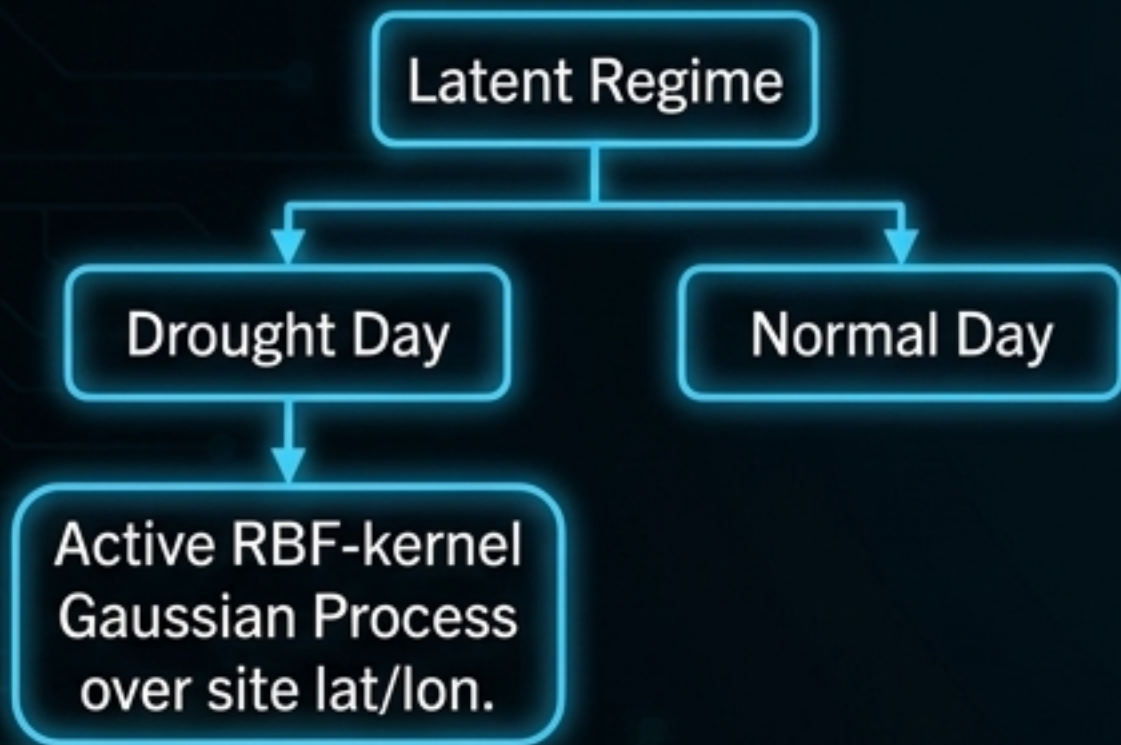
The Methodological Contenders

Rule-of-Thumb	Academic Control	Real ISO Practice	Advanced Baseline	The Challenger
• ERCOT N-1: Real rule sizing against the single largest unit. • Generic 5% wind rule.	• Independence (Method 1): Draws each site independently without spatial correlation.	• Aggregate Correlation (Method 2): Uses standard single-factor equicorrelation sampler.	• Vanilla Spatial GP (Method 3): Shared spatial covariance fit via marginal likelihood.	• GP + soft-EM (Method 4): 2-component Gaussian mixture on fleet-wide deviation.

Phase 0: Validating the Oracle's Tail Dependence

Generative Process

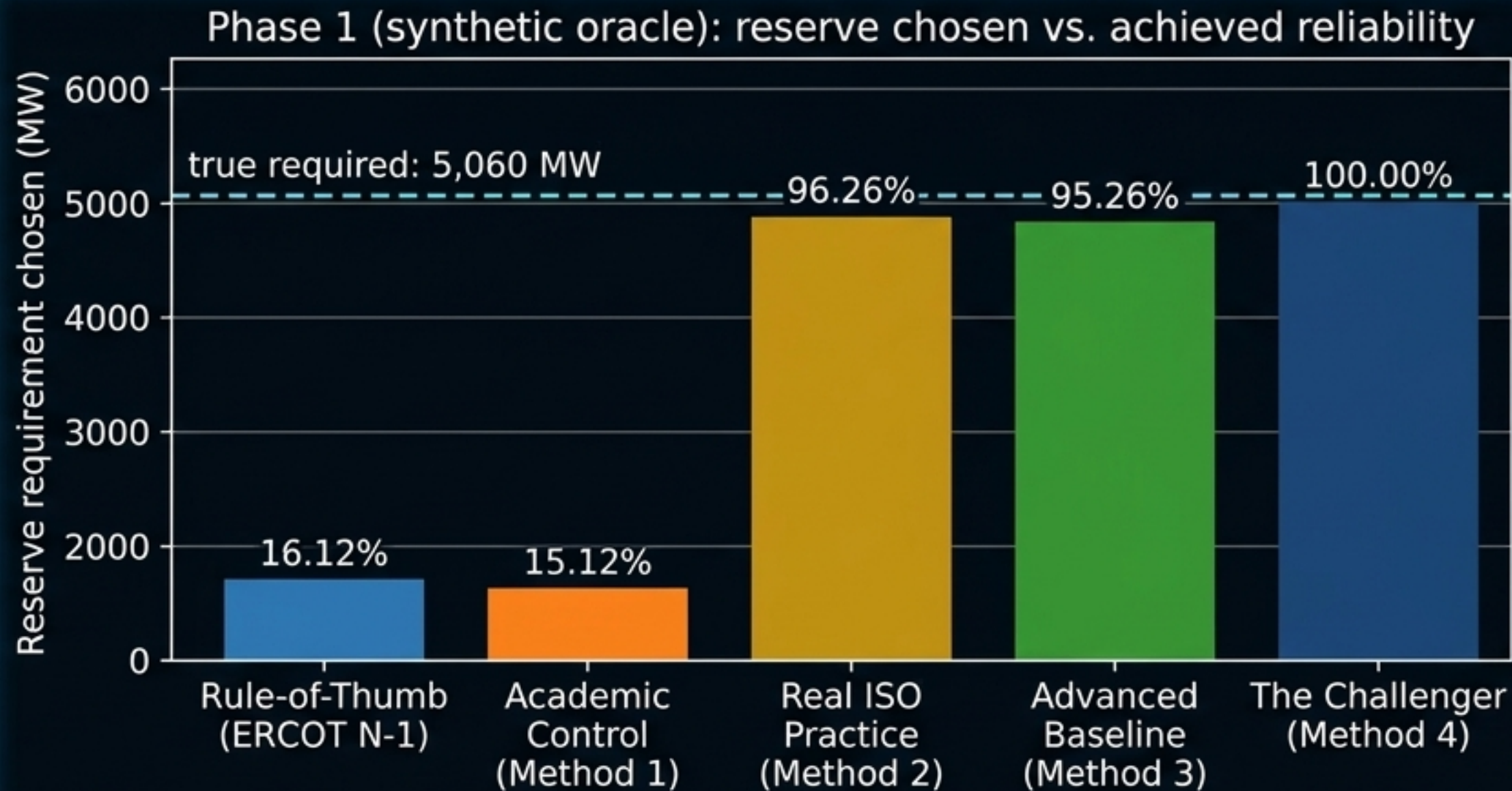
To test the methods, the lab built a synthetic world with a known ground truth. It uses a two-layer generative process per day:



Crucial Insight: The oracle has genuine tail dependence (0.607 for near pairs). A Gaussian model fit to the exact same mean and covariance gets this badly wrong (0.082). It cannot reproduce joint-tail risk.

Phase 1 Synthetic Reliability: Only Method 4 Clears the Target

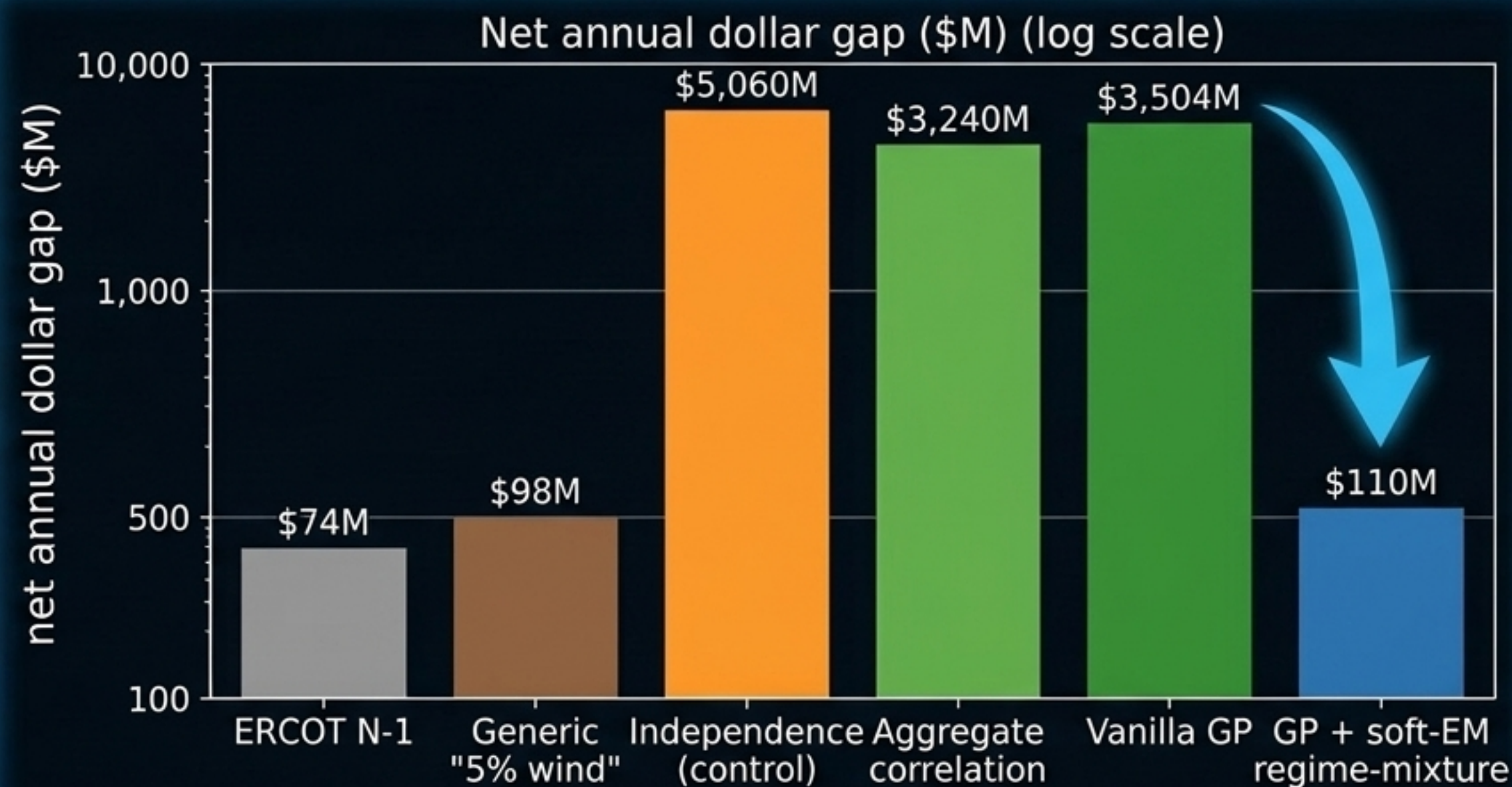
Scored against 100 synthetic sites | 500,000-day scoring sample | True required reserve: 5,060 MW



- Every traditional method, including Real ISO Practice (Aggregate Correlation, 96.26%), severely under-reserves against the tail risk.
- Method 4 (GP + soft-EM) achieves 100.00% reliability, safely clearing the 99.9726% target line.

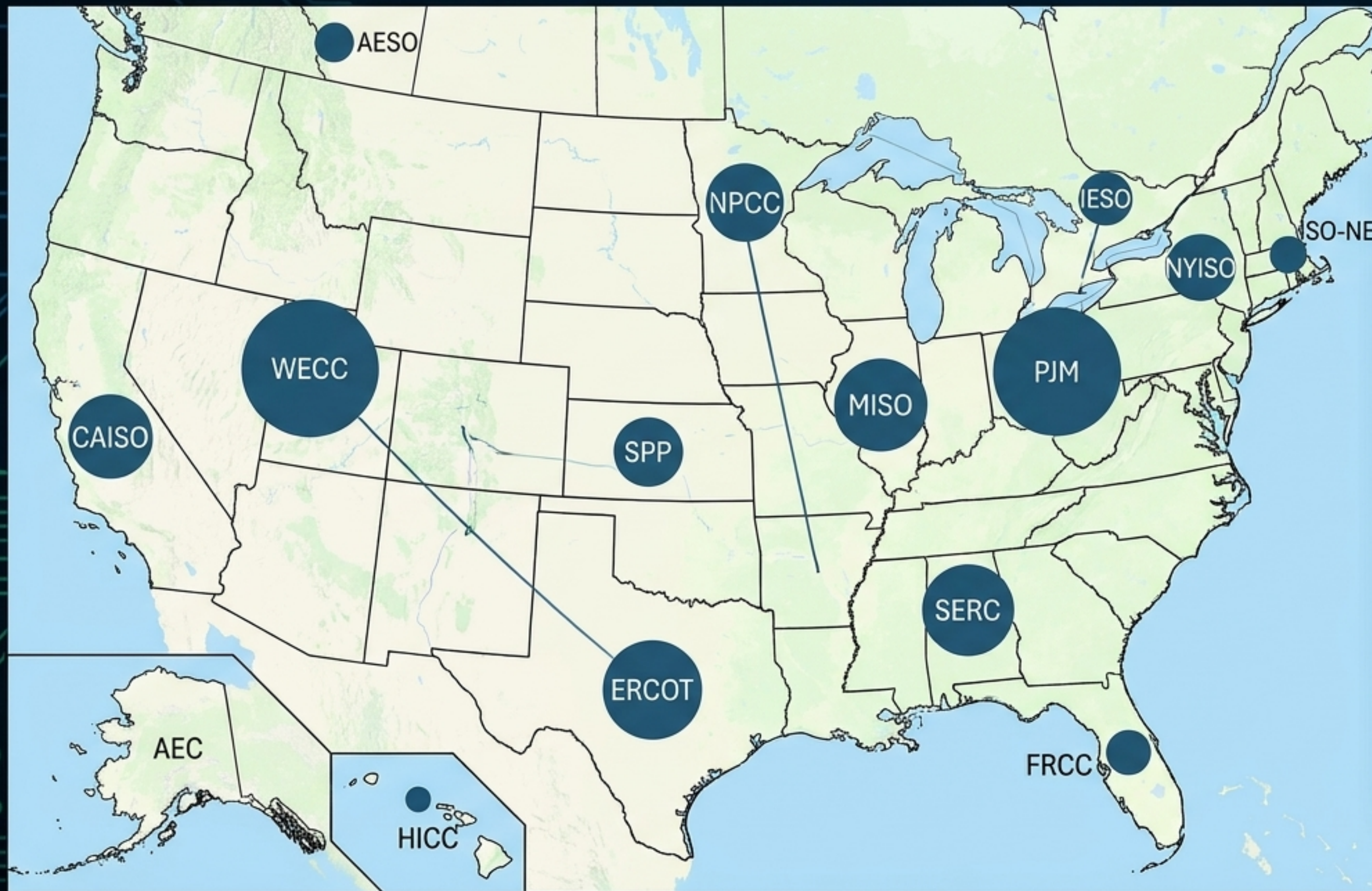
Phase 1 Cost: The Billion-Dollar Penalty for Missing the Tail

Scored against 100 synthetic sites | 500,000-day scoring sample | True required reserve: 5,060 MW



Why does Vanilla Spatial GP fail to beat Aggregate Correlation? This is a verified finding, not a bug. A smooth spatial kernel alone is insufficient to capture the risk. Representing the regime itself is what closes the gap, lowering the penalty to just \$110M for Method 4 compared to \$3.5B for Method 3.

Phase 2: Moving to the Real-World Fleet



Data Source:

Real EIA-930 hourly generation-by-fuel-type data.

Scale:

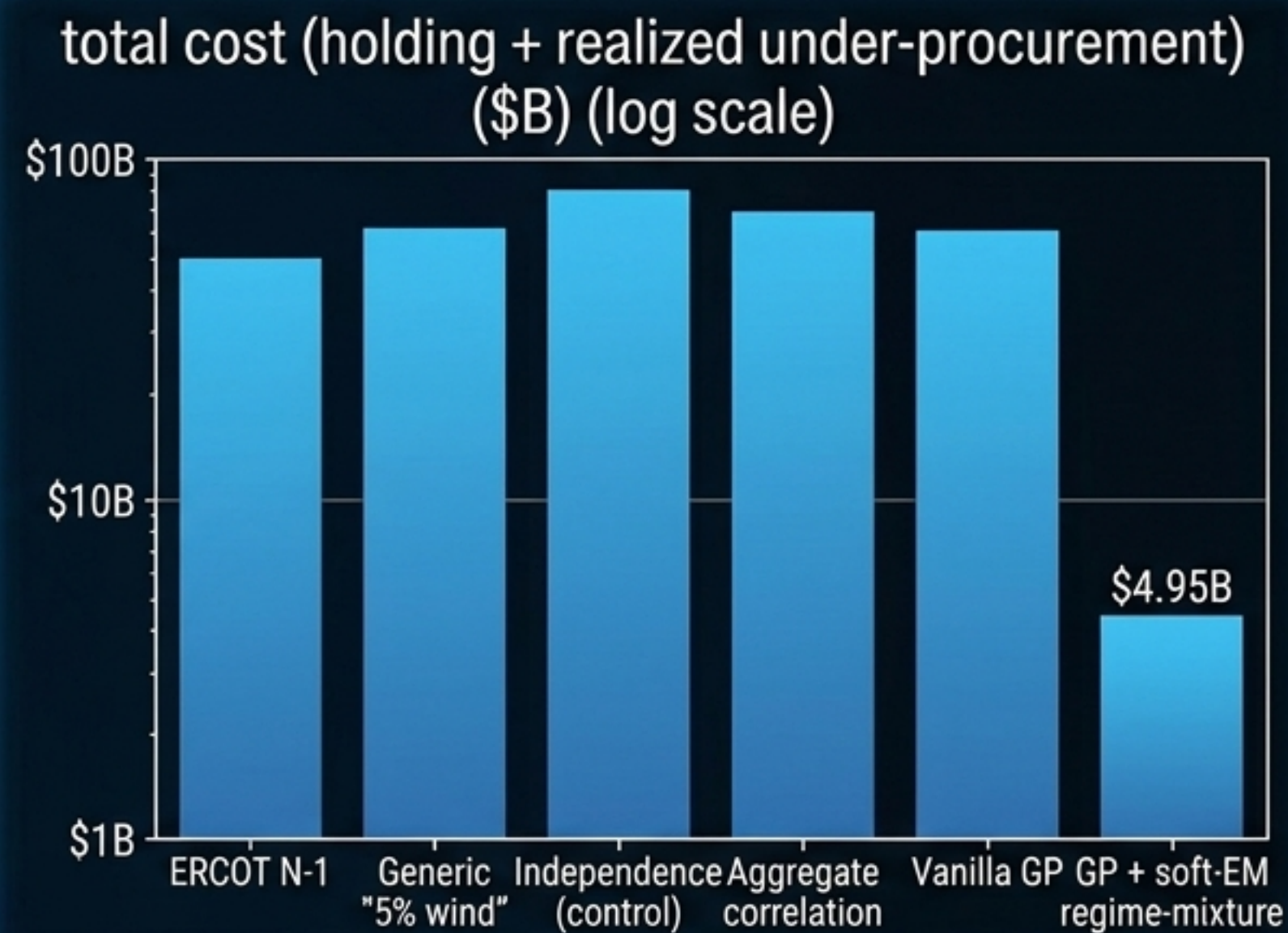
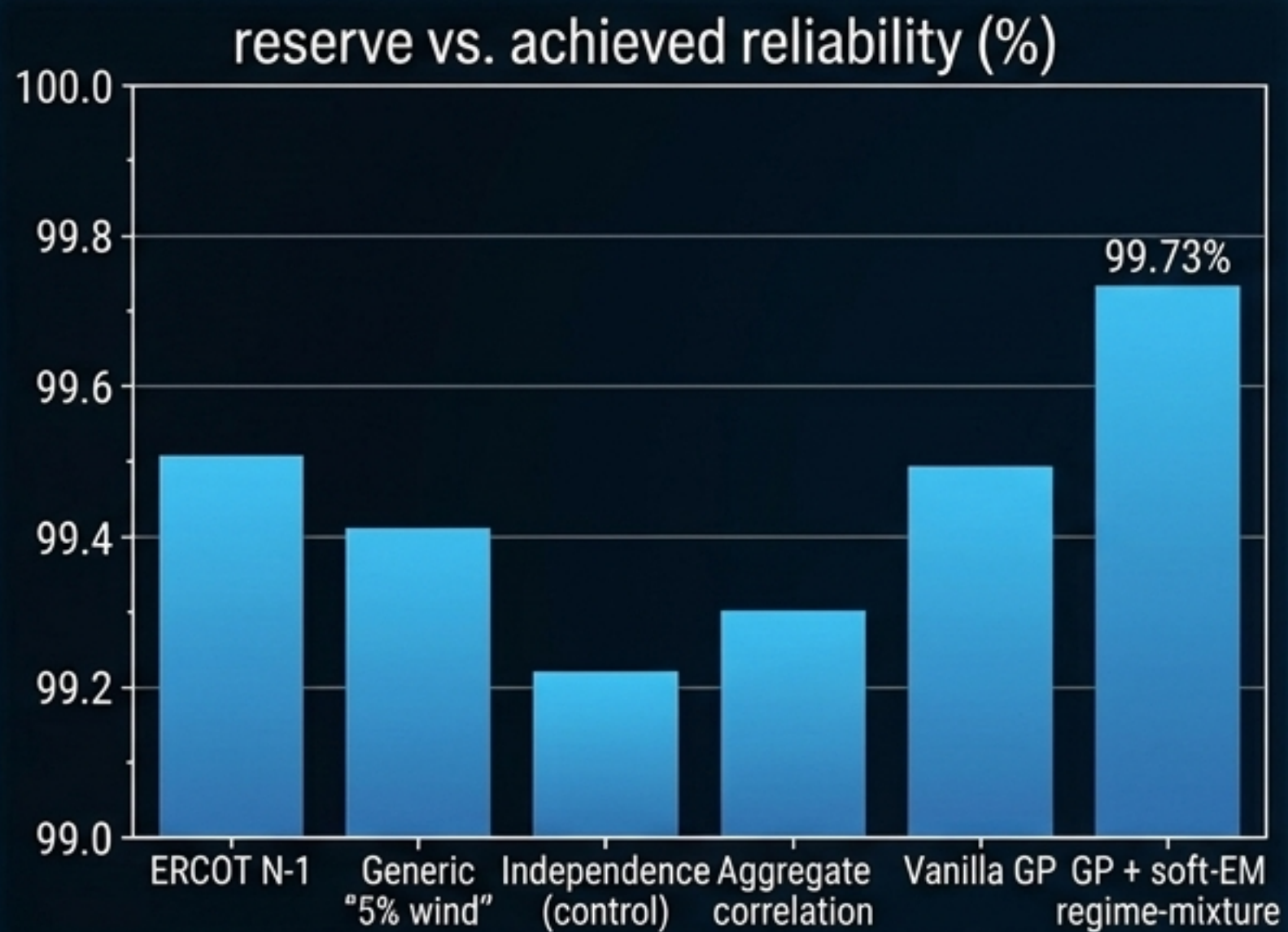
15 real US Balancing Authorities (treated as individual “sites” at territory centroids).

Validation Structure:

Fit entirely on 2023 data.

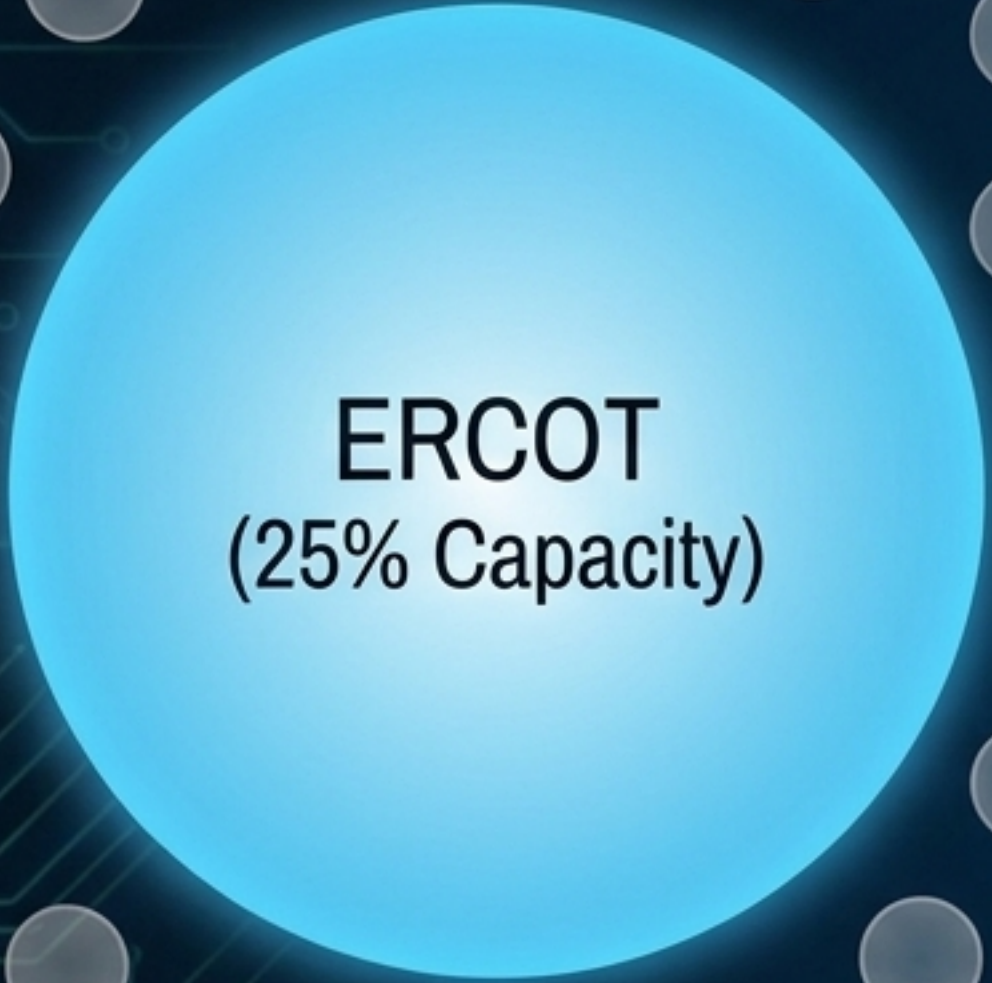
Scored completely out-of-sample on held-out 2024 data.

Phase 2 Final Scorecard: Maximum Reliability at the Lowest Cost



The identical qualitative ranking from the synthetic oracle survived contact with real, messy EIA-930 data. GP + soft-EM regime-mixture decisively outperforms the aggregate-correlation baseline used by real ISOs.

Nuance: The ERCOT N-1 ‘Success’ is a Resolution Artifact



ERCOT
(25% Capacity)

The Anomaly

ERCOT's real N-1 rule scored a surprisingly high 99.18% reliability in Phase 2. Is the rule secretly sound for correlated-shortfall risk?

The Reality

No. In a 15-site sample of Balancing Authorities, ERCOT alone makes up 25% of total fleet nameplate capacity. The single-largest-unit rule coincidentally covers a lot of system variability because the “unit” in this resolution is massive.

Phase 1's 100-site synthetic fleet (where no site dominates) proved N-1 actually performs poorly for system-wide shortfalls.

Fraser's Question: Does Method 4 Win Because It Doesn't 'Throw Away Data'?

The lab ran the identical regime-detection GMM through two downstream GP-fitting schemes to isolate the exact mechanism of victory.



The Answer is Conditional on Regime Imbalance

Synthetic Data (RARE Regime)

~5% probability
(34 hard-partitioned
drought days out of 730).



Result: Soft is highly superior.
Soft = \$109.9M vs Hard = \$121.2M net gap.

Mechanism: Hard-partitioning starves the drought-component GP of data. Soft allows all 730 days to contribute, rescuing the minority component.

Real Data (BALANCED Regime)

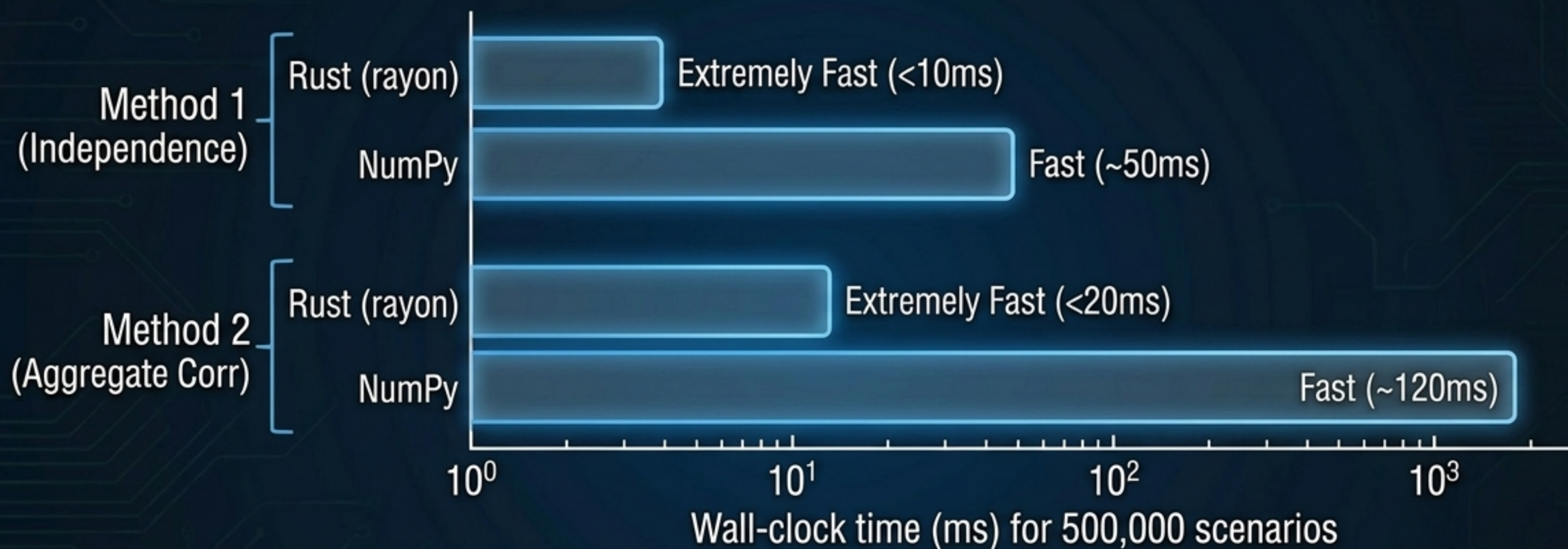
~50% split
(180 hard-partitioned
drought days out of 365).



Result: Statistical Tie.
Soft = \$4.95B vs Hard = \$4.92B total cost.

Mechanism: When data is plentiful for both regimes, the soft/hard distinction vanishes. The true advantage is representing a regime-mixture at all.

Computational Fairness: Traditional Methods Are Fast, Just Mathematically Inadequate



The Insight: The traditional baselines were implemented in a highly optimized Rust crate. Their massive billion-dollar losses are not an artifact of being handicapped on computational speed—they are fundamentally mathematically incapable of modeling genuine tail risk.

Final Lab Synthesis

Genuine Tail Dependence Required	GP + Soft-EM Wins
The synthetic oracle proved that Gaussian models with matching mean and covariance systematically fail to reproduce true tail risk, leaving a 7.4x magnitude gap.	The qualitative ranking holds across both synthetic oracles and held-out real EIA-930 data. Method 4 decisively achieves the highest reliability at the lowest total cost.
Conditional Value of Data Retention	Grounded in Reality
“Not throwing away data” (Soft EM) is a highly valuable mechanism for rescuing data-starved rare regimes, but proves statistically irrelevant for balanced, 50/50 splits.	Every dollar figure relies on sourced constants (VOLL, NERC conventions, capacity costs). Non-obvious modeling choices were driven by real diagnostic trails, not upfront design.